ADVANCED QUANTUM MECHANICS

Exercise 3 – presentation session – 28.10.2025

TECHNISCHE UNIVERSITÄT DARMSTADT

Claudiu Genes, Institute for Applied Physics; TU Darmstadt, Winter semester 2025/2026

We'll have three presentations — one per exercise. The tutor will choose three volunteers and randomly assign one exercise to each.

1. Time ordering

Let's understand the need for time ordering. Take the Schrödinger equation

$$\mathrm{i}\hbar\,\partial_t\,|\psi(t)\rangle = \hat{H}(t)\;|\psi(t)\rangle\,.$$

(i) Turn it into a difference equation for a small time interval Δt such that

$$|\psi(\Delta t)\rangle \approx \left(\mathbb{1} - \frac{\mathrm{i}}{\hbar}\,\hat{H}(0)\,\Delta t\right)|\psi(0)\rangle\,.$$

(ii) Use the notation $H(j\Delta t) = H_j$ and write the difference equations for the next two terms $|\psi(2\Delta t)\rangle$ and $|\psi(3\Delta t)\rangle$.

(iii) Arrange the terms in $|\psi(3\Delta t)\rangle$ to get:

$$|\psi(3\Delta t)\rangle = \left[\mathbb{1} - \frac{\mathrm{i}\,\Delta t}{\hbar}\,(H_2 + H_1 + H_0) - \frac{(\Delta t)^2}{\hbar^2}\,(H_2H_1 + H_2H_0 + H_1H_0) + \frac{\mathrm{i}\,(\Delta t)^3}{\hbar^3}\,H_2H_1H_0\right]|\psi(0)\rangle\,,$$

showing how the Hamiltonians are always ordered in time.

(iv) For a time-independent Hamiltonian H, use the fact that all $H_j = H$ and generalize the solution above from 3 to any number N such that $t = N\Delta t$. Use some binomial expansions and the following limit

$$\lim_{N\to\infty}\left(1 + \frac{\hat{A}}{N}\right)^N = e^{\hat{A}}$$

(meaning that one takes the limit $\Delta t \to 0$) to find the solution $|\psi(t)\rangle$ that you expected to find.

2. Solving the Schrödinger equation in the position representation

Consider a particle coming from the left that bounces off or tunnels through a delta-function *repulsive* potential. We will not solve the time evolution but only the "asymptotic" behavior (using plane waves): this means we only solve the stationary Schrödinger equation

$$-\frac{\hbar^2}{2m}\,\frac{d^2\psi(x)}{dx^2} + \lambda\,\delta(x)\,\psi(x) = E\,\psi(x),$$

where

$$V(x) = \lambda\,\delta(x), \qquad \lambda > 0.$$

Assume that for $x < 0$ (to the left of the scatterer)

$$\psi_{\mathrm{L}}(x) = e^{ikx} + r\, e^{-ikx},$$

where e^{ikx} is the incoming wave and re^{-ikx} is the reflected wave. For $x > 0$ (to the right of the scatterer) we have only a transmitted wave,

$$\psi_{\mathrm{R}}(x) = t\, e^{ikx}.$$

(i) Write the continuity equation at the origin $\psi(0^+) = \psi(0^-)$ and find the connection between r and t (condition 1).

(ii) Integrate Schrödinger's equation from $-\varepsilon$ to $+\varepsilon$ and let $\varepsilon \to 0^+$. This gives a condition for the jump of the derivative of the wavefunction over the delta potential $\psi'(0^+) - \psi'(0^-) = \frac{2m\lambda}{\hbar^2}\,\psi(0)$ at the origin (condition 2).

(iii) From the two conditions, find the complex reflection and transmission amplitudes (set $\eta = \frac{m\lambda}{\hbar^2 k}$):

$$t = \frac{1}{1 + i\eta}, \qquad r = \frac{-i\eta}{1 + i\eta}, \qquad t = 1 + r$$

and the reflection and transmission probabilities defined as $R = |r|^2$ and $T = |t|^2$.

(iv) Discuss the cases of low energy ($k \to 0$) and high energy ($k \to \infty$) and check if your intuition is validated.

This result will be useful later, when we introduce scattering theory. We will re-derive these coefficients using the Lippmann–Schwinger formalism as a concrete example.

3. The quantum harmonic oscillator in a thermal state

Let's assume a particle in a harmonic potential and in a thermal state (for example a vibrating mirror in a room at temperature T). The thermal state is described by a density operator

$$\hat{\rho}_{\mathrm{th}} = \sum_{n=0}^{\infty} \left(1 - e^{-\beta\hbar\omega}\right) e^{-\beta\hbar\omega n} \left|n\right\rangle \left\langle n\right|, \qquad \beta = \frac{1}{k_B T}.$$

(i) Compute the expectation value of the position operator

$$\langle \hat{x} \rangle = \mathrm{Tr}[\hat{\rho}\, \hat{x}] = \sum_{n=0}^{\infty} \left\langle n\right| \hat{\rho}\, \hat{x} \left|n\right\rangle$$

from

$$\hat{x} = x_{\mathrm{zpf}}(\hat{a} + \hat{a}^\dagger), \qquad x_{\mathrm{zpm}} = \sqrt{\frac{\hbar}{2m\omega}}.$$

(ii) Show that the average excitation number

$$\bar{n} = \mathrm{Tr}[\hat{\rho}\, \hat{n}] = \sum_{n=0}^{\infty} \left\langle n\right| \hat{\rho}\, \hat{n} \left|n\right\rangle$$

assumes the expression

$$\bar{n} = \frac{1}{e^{\beta\hbar\omega} - 1}.$$

(iii) Show that the density operator can be written as

$$\hat{\rho}_{\mathrm{th}} = \sum_{n=0}^{\infty} \frac{\bar{n}^n}{(1 + \bar{n})^{n+1}} \left|n\right\rangle \left\langle n\right|,$$

and write the density operator in the zero- and infinite-temperature limits.